

Databusters

NUS Data Science and Economics Student Committee
NTU Economics and Data Science Sub-Club

January 27, 2026

Welcome to Databusters! The objective of this datathon is to test your ability to apply data analysis to real-world economics problems. This year’s case examines financial runs across two crises: the 2008 Reserve Primary Fund collapse and the 2022 Terra-Luna stablecoin failure.

1 Problem Statement

1.1 Background

Financial stability relies on the shared confidence that investors can redeem their assets at face value on demand. When this confidence weakens, financial systems that might appear safe can experience sudden and self-reinforcing runs [1]. Traditionally, this risk of financial crisis has been associated with banks and money market funds (MMFs).

More recently, a “new age” financial system that claims stability has emerged: stablecoins. Stablecoins are a type of cryptocurrencies designed to maintain a fixed value, usually 1 USD. Like MMFs, stablecoins are marketed as low-risk, highly liquid vehicles for storing value and facilitating payments. In both stablecoins and MMFs, users expect stability, immediacy, and near-riskless redemption [2].

Algorithm-based stablecoins are frequently viewed as fundamentally different from MMFs: algorithm-based stablecoins rely on automated rules and blockchain technology rather than regulated intermediaries. Yet recent events suggest that stablecoins may be subject to the same “traditional” financial stability problems [3].

1.2 The “traditional” financial crisis

In September 2008, the Reserve Primary Fund experienced a run after exposure to Lehman Brothers debt, where its net asset value fell below the standard \$1 mark. This event triggered approximately \$40 billion in redemption requests within days, forcing the Federal Reserve and U.S. Treasury to implement emergency interventions to prevent a broader collapse of the money market fund industry [4].

1.3 The “new age” financial crisis

A similar breakdown occurred in May 2022 within the crypto ecosystem with TerraUSD (UST), an algorithmic stablecoin. UST maintained its dollar peg through an automated arbitrage mechanism with its sister token, LUNA [3]. When confidence in the peg weakened, mass redemptions triggered a rapid feedback loop, commonly described as a “death spiral.” Within 72 hours, LUNA’s market value effectively collapsed, wiping out nearly \$50 billion in value. Unlike in 2008, this crisis unfolded without regulatory oversight, deposit insurance, or a lender of last resort. Instead, all transactions were recorded in real time on a public blockchain.

These two events highlight a shared vulnerability: during periods of high stress, assets that promise immediate liquidity can become difficult to redeem at face value. Loss of confidence leads investors to rush for exit, causing prices of MMFs and stablecoins to fall below their promised pegs and triggering self-reinforcing runs.

2 Your Task

You are a research team studying financial run dynamics. Your analysis must combine data visualization, economic reasoning, and policy thinking.

Investigate whether runs exhibit similar economic dynamics across two radically different institutional settings: the collapse of the Terra-Luna stablecoin system in 2022 and the run on the Reserve Primary Fund during the 2008 Global Financial Crisis. Using Terra's on-chain data and money market fund indicators from 2008, analyze how stress emerges, how behavior changes as confidence deteriorates, and how institutional design shapes outcomes.

2.1 Section A: Run Dynamics

Select at least two questions from this section. Your answer should be supported using data.

1. When the Peg Breaks: Onset and Spread

Identify when confidence begins to break down in the Terra-Luna stablecoin run (2022) and the Reserve Primary Fund run (2008). How does stress first become visible, and how does panic propagate across participants in each case?

2. Where Does the Money Go?

For each of the runs, where does capital move? What do these reallocations reveal about which assets investors trust during periods of financial instability?

3. Who Bears the Losses?

Identify which parties suffer losses in each crisis. Explain why the system could not protect them and how differences in design made the outcomes more or less severe.

2.2 Section B: Designing for Confidence

You must attempt this section.

Drawing on your findings from Section A, propose one policy or design change that could reduce the likelihood or severity of a stablecoin run like Terra-Luna. Explain how your recommendation would prevent the specific failure you identified, and discuss the trade-offs involved.

3 Data

The stablecoin on-chain data can be accessed and downloaded via the `ERC20-stablecoins.zip` file provided.

Data related to the 2008 Global Financial Crisis, including money market fund flows and macro-financial indicators, are provided in the `gfc.zip` file.

In addition to the provided datasets, you may use any publicly available data. All sources must be cited.

4 Deliverables

4.1 Slide Deck

- Submit as a **PDF file**
- Maximum of **10** slides excluding title and references
- Group name and member details should appear on the **title page only** and must not be included elsewhere in the submission

Your slides should present:

- Key visualizations comparing crisis dynamics across both episodes
- Structured economic reasoning explaining what the data reveals
- Institutional comparison based on quantitative analysis
- Policy evaluation

4.2 Technical Appendix (Mandatory, Separate File)

- Reproducible code (Python or R)
- Data loading, cleaning, and analysis
- Calculations and visualizations

We will execute your code. Non reproducible code will be penalized.

5 Submission and Timeline

Submission Deadline: Sunday, February 8, 2026 at 11:59 PM (SGT)

Naming Convention: Databusters26-[Group Name]-[Report/Code]

Submit your deliverables via email to:

- `pd.databusters.dsesc@u.nus.edu`
- `sssclub-ecds-chair@e.ntu.edu.sg`

The top 5 teams will be selected based on:

- Economic Reasoning
- Data Analysis and Visualization
- Clarity and Storytelling
- Originality

Announcement of Finalists: Thursday, February 19, 2026

Final Presentations: Saturday, February 21, 2026, 9:30 AM–1:00 PM

Each team will have a maximum of 10 minutes to present.

References

- [1] Gorton, G. (1988). Banking Panics and Business Cycles. *Oxford Economic Papers*, 40(4), 751–781.
- [2] Anadu, K., Azar, P., Cipriani, M., et al. (2025). Runs and Flights to Safety: Are Stablecoins the New Money Market Funds? *FRB of Boston Supervisory Research & Analysis Unit Working Paper No. SRA 23-02*.
- [3] Liu, J., Makarov, I., & Schoar, A. (2023). Anatomy of a Run: The Terra Luna Crash. NBER Working Papers 31160, National Bureau of Economic Research, Inc.
- [4] Gorton, G., & Metrick, A. (2010). Regulating the Shadow Banking System.